ModelIt!

Teacher's Guide

Why Can't You See in the Dark?

The Surprising Science of Light and Vision

Quick Reference

Grade Level	Grade 4
NGSS Standard	4-PS4-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 4-PS4-2

Develop a model to describe that light reflecting from objects and entering the eye allows objects to be seen.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how light source brightness, light reflection, and eye detection work together in the vision system.
Disciplinary Core Idea	**PS4.B Electromagnetic Radiation** — An object can be seen when light reflected from its surface enters the eyes. Objects can be seen only when light is available to illuminate them.
Crosscutting Concept	**Cause and Effect** — Students explore how the presence or absence of light (cause) directly determines whether objects can be seen (effect).

CAST Testing Alignment

MULTIPLE CHOICE:

A student walks into a dark closet and cannot see anything, even though the closet is full of clothes. Which best explains why the student cannot see?

CONSTRUCTED RESPONSE:

Explain how you are able to see this paper right now. Trace the path of light from where it starts to how it reaches your eyes. Use the words 'light source,' 'reflection,' and 'eye' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Light is the reason we can see the beautiful world around us. Without light, vision is impossible — no matter how good your eyes are. Understanding how light, objects, and eyes work together is one of the most important ideas in physical science.

Light Sources vs. Reflectors

There are two kinds of objects in the world: those that make their own light (luminous objects) and those that reflect light made by something else (illuminated objects). The Sun, light bulbs, phone screens, and fire are all light sources. Everything else — books, tables, people, the Moon — is visible only because light bounces off them. The Moon looks bright, but it is actually just reflecting sunlight!

How Reflection Works

When light hits an object, some of it bounces off (reflects), some passes through (transmits), and some is absorbed (turned into heat). Smooth, shiny surfaces like mirrors reflect light in an organized way — this is called specular reflection. Rough surfaces like paper scatter light in many directions — this is called diffuse reflection. Both types of reflection allow us to see objects, but mirrors create clear images while rough surfaces just look lit up.

How the Eye Works

Light enters the eye through the cornea (clear front surface), passes through the pupil (the opening that controls how much light gets in), and is focused by the lens onto the retina at the back of the eye. The retina has millions of special cells called rods and cones that detect light and color. These cells send electrical signals through the optic nerve to the brain, which assembles the signals into the images you see.

Why Darkness Means Invisibility

In complete darkness, there is no light to bounce off objects and enter your eyes. Your retina has nothing to detect, so no signals are sent to your brain. This is why you cannot see anything in a pitch-black room, no matter how long you wait. Your eyes need incoming light to create vision — they do not produce their own light to scan the room, despite what some ancient Greeks believed!

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify light source brightness as an external component and light bouncing off objects and eye detection as internal components that respond to the amount of available light.

E **Establish Relationships**
Students determine that light source brightness positively affects light bouncing off objects, and more reflected light positively affects eye detection — a clear cause-and-effect chain.

V **Visualize & Model**
Students build a model showing how brighter light leads to more reflection, which leads to better vision, and how removing light eliminates vision entirely.

E **Evaluate Outcomes**
Students run pitch black, dim flashlight, and bright sunny day scenarios to observe how changing brightness affects the whole vision system.

R **Revise & Extend**
Students add object color or pupil size to explore why some objects are easier to see than others.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Close your eyes tight. What do you see? Nothing! Now open them. What changed? The same room was always there."*

Watch For

Have students close their eyes for 5 seconds, then open. Discuss: the room did not change, but your ability to see it did.

Slide 2: Learning Objectives

3 min

Say: *"Today we are going to solve the mystery of WHY darkness makes things invisible."*

Watch For

Have students read objectives aloud. Pre-teach 'reflection' by bouncing a flashlight off a mirror.

Slide 3: Big Question

3 min

Say: *"Imagine a room full of your favorite things — toys, games, snacks — but it is completely dark. They are ALL there. Why can you not see a single one?"*

Watch For

Have students turn and talk: What does your eye actually need to see something?

Slide 4: LEVER Framework

2 min

Say: *"We are going to build a model of light and vision to figure this out."*

Watch For

Preview each LEVER step. Ask: Do you think your eyes send out something, or take something in?

Slide 5: Activity 1: Components

8 min

Say: *"What are the key parts of the seeing system? What has to happen for you to see a book?"*

Watch For

Guide sorting. Explain that light source brightness is external because it comes from outside the vision system.

Slide 6: Activity 2: Connections

8 min

Say: *"When the light gets brighter, does MORE or LESS light bounce off the book? And what happens in your eye?"*

Watch For

Help students draw positive arrows through the chain: brightness to reflection to eye detection.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let us test it! What happens when we take ALL the light away?"*

Watch For

Students predict first. Run pitch black, dim, and bright scenarios. The pitch black result is the key 'aha' moment.

Slide 8: Discoveries

5 min

Say: *"Now you know — your eyes do not make light, they COLLECT light! No light means no vision."*

Watch For

Draw the full path on the board: light source to object to eye to brain. Emphasize that the eye is a receiver, not a sender.

Slide 9: STEM Challenge

12 min

Say: *"Time to become light engineers! Can you bounce light through a maze using mirrors?"*

Watch For

Distribute materials (small mirrors, flashlights, cardboard). Groups build light mazes and guide a beam to a target.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Light Source Brightness
- **INTERNAL:** Light Bouncing Off Objects, Eye Detection

Relationships

- **Light Source Brightness to Light Bouncing Off Objects:** POSITIVE (+) — Brighter light means more light rays hit objects and more light bounces off them. On a sunny day, everything looks vivid because tons of light is reflecting everywhere.
- **Light Bouncing Off Objects to Eye Detection:** POSITIVE (+) — More reflected light entering the eye means the retina detects a stronger signal. This is why objects are easy to see in bright light and hard to see in dim light — your eyes need reflected light to work.

Key Discoveries

- You can ONLY see an object when light bounces off it and enters your eyes — no light means no vision
- Light sources create light, but most objects just REFLECT light — they do not make their own
- Brighter light means more light bounces off objects, which means better eye detection
- Your eyes do not shoot out beams to see things — they RECEIVE light that comes in from outside

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Your eyes send out invisible beams to see things"	Your eyes do NOT send out anything. They are receivers, like a satellite dish. Light from a source hits an object, bounces off, and enters your eye. The eye collects incoming light — it does not project outgoing beams. Ancient Greeks called this the 'emission theory' and it is wrong!
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"You can see in total darkness if you wait long enough"	No matter how long you wait, you cannot see in total darkness. Your eyes need light to function. In dim light, your pupils get bigger to collect more of the available light, which is why your eyes 'adjust' — but if there is truly ZERO light, there is nothing to adjust to.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"The Moon makes its own light"	The Moon is a rocky object that does not produce any light at all. It appears bright because sunlight hits its surface and reflects toward Earth. During a lunar eclipse, when Earth blocks the sunlight from reaching the Moon, the Moon goes dark — proving it needs the Sun's light to be visible.